

Data communications (Lab 9): Frequency Domain Representation of a Sine Wave

Objective: To extract the amplitude and frequency components from a user-defined sine wave equation and plot its frequency domain representation in MATLAB.

Theory

A sinusoidal signal is generally written as: $x(t) = A\sin(2\pi ft)$

where, $A = \text{Amplitude}$

$f = \text{frequency}$

$t = \text{time}$

For a combination of multiple sine waves:

$$x(t) = A_1\sin(2\pi f_1 t) + A_2\sin(2\pi f_2 t) + A_3\sin(2\pi f_3 t) \dots$$

Each sine wave contributes one frequency components.

Example: $x(t) = 5\sin(2\pi 50t) + 3\sin(2\pi 120t)$

Frequency components are: (Amplitude 5; Frequency: 50Hz), (Amplitude 3, Frequency: 120 Hz)

In the frequency-domain representation:

x-axis→Frequency

y-axis→Amplitude

Each component shows as a vertical line at its corresponding frequency.

```
clear;
clc;
close all;

% Write equation as text
eqn = '5*sin(2*pi*50*t) + 3*sin(2*pi*120*t)';

% Extract amplitude and frequency
tokens = regexp(eqn, '(\d+)\*sin\((2\*pi\*(\d+)\*t\))', 'tokens');

A = [];
f = [];

for i = 1:length(tokens)
    A(i) = str2double(tokens{i}{1}); % amplitude
    f(i) = str2double(tokens{i}{2}); % frequency
end

% Plot frequency spectrum
stem(f,A,'filled','LineWidth',2)

grid on
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Frequency Domain Representation')
```

```
% Set axis limits
axis([0 max(f)+10 0 max(A)+1])
```

Code Explanation

1. Initialization

```
clc;
clear;
close all;
```

- `clc` → Clears command window
- `clear` → Removes all variables

2. Store Equation as Text

```
eqn = '5*sin(2*pi*50*t) + 3*sin(2*pi*120*t)';
```

Here the sine equation is written as a **string**.

3. Extract Amplitude and Frequency

```
tokens = regexp(eqn, '(\d+)\*sin\((2\*pi\*(\d+)\*t)\)', 'tokens');
```

This is the main extraction step.

`regexp` means **Regular Expression**.

It searches the equation and finds patterns matching:

```
number * sin(2*pi*number*t)
```

Pattern:

```
(\d+)\*sin\((2\*pi\*(\d+)\*t\)
```

Breakdown:

First `(\d+)` → Extracts amplitude.

Second `(\d+)` → Extracts frequency.

For:

```
5*sin(2*pi*50*t)+3*sin(2*pi*120*t)
```

MATLAB finds:

```
tokens =
{
    {'5','50'}
    {'3','120'}
}
```

4. Create Empty Arrays

```
A = [];
f = [];
```

These arrays store:

- amplitudes → `A`
- frequencies → `f`

5. Store Extracted Values

```
for i = 1:length(tokens)
```

Loop runs through each extracted sine term. Since there are 2 sine waves: Loop runs **2 times**.

First iteration:

```

A(i) = str2double(tokens{i}{1});
takes:
tokens{1}{1}='5' → Converts string → number: A(1)=5
Then:
f(i) = str2double(tokens{i}{2});
takes:
tokens{1}{2}='50' → Converts string → number : f(1)=50
Now:
A = [5]
f = [50]

```

Second iteration:

Extracts:

```

tokens{2}{1}='3'
tokens{2}{2}='120'

```

Stores:

```

A = [5 3]
f = [50 120]

```

Final arrays:

```

Amplitude: A = [5 3]
Frequency: f = [50 120]

```

6. Plot Frequency Spectrum

```

stem(f,A,'filled','LineWidth',2)
stem() draws vertical lines.
x-values: f = [50 120]
y-values: A = [5 3]

```

So MATLAB draws:

- line at **50 Hz**, height **5**
- line at **120 Hz**, height **3**

This is the frequency-domain representation.

7. Add Grid and Labels

```

grid on
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Frequency Domain Representation')

```

8. Set Axis Limits

```

axis([0 max(f)+10 0 max(A)+1])

```

Conclusion: This code automatically extracts amplitude and frequency from a sine equation using regular expressions, stores them in arrays, and plots the line spectrum in frequency domain using stem().